

IntegralAI — AI-Based Integration Learning Platform

LAYER 1: USER / CLIENT LAYER

Student

Role & Auth

Administrator

LAYER 2: PRESENTATION LAYER — React / Next.js Frontend

Auth UI

• Login / Register

Student UI Views

• Dashboard & Solver UI

• Step Solution & Explanation

Practice UI

• Practice Mode

• Similar Problems

Admin UI Views

• Admin Dashboard

• User & Question Bank

Frontend Services

• API Client (Axios/Fetch)

• State Validation

LAYER 3: APPLICATION LAYER — Backend REST API / Node.js & Express.js

REST API Router & Middleware Boundary

Auth Module

Controller & Service

User Module

Profile & Roles

History Module

Save / Fetch Queries

Practice Module

Question Retrieval

Admin Module

User & Bank Mgmt

Problem & Solver Module

• Problem Controller

• Solver Service Bridge

Progress Module

Stats & Topic Mastery

POST /api/solve

LAYER 4: INTEGRATION SOLVER & AI PROCESSING PIPELINE

1. Input Processing

Expression Normalization

2. Problem Analysis

3. Method Selection

Stated in RAG document

4. Math Solver

• Apply Selected Formula

• Using Qwen LLM

5. SymPy Verification

Candidate

Computer Algebra System

Validates Candidate Result

6. Step-by-Step Generator

Structures Logical Flow

7. AI Explanation Service

LLM / AI Pedagogical Feedback

External AI / LLM Provider

OpenAI, QWEN LLM

LAYER 5: DATA LAYER — Vector Database

Users Coll.

Profiles, Auth, Roles

History Coll.

User Problem Logs

Progress Coll.

Mastery & Metrics

Problems Coll.

Solutions Coll.

Steps & SymPy Proof

Question Bank

Curated Practice

LAYER 6: DEPLOYMENT & INFRASTRUCTURE LAYER — Containerized Environment (Docker)

Frontend Container

REACT

Backend & Solver Container

Express.js + Python/SymPy Runtime

Database Container

Vector Database